Submodular Optimization and Greedy Approximation for Distributed Influence Maximization

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Project Goal

- Topic 3: Influence Maximization.
- Choose k seed nodes in a network to maximize expected diffusion spread.
- Connect a modern network data problem to classical algorithms:
  - monotone submodular maximization;
  - greedy approximation;
  - CELF lazy evaluation;
  - GreeDI-style distributed candidate sharing.

Problem Formulation

$$\max_{S \subseteq V,\ |S| \leq k} \sigma(S)$$

- $G = (V, E, p)$ is a directed graph with edge activation probability p_{uv}.
- $\sigma(S)$ is the expected number of active nodes after diffusion.
- We use the Independent Cascade model:
  - newly active node u gets one chance to activate each inactive out-neighbor v;
  - activation succeeds independently with probability p_{uv}.

Algorithmic Foundation

- Under IC diffusion, $\sigma(S)$ is monotone and submodular.
$$\sigma(A \cup \{v\}) - \sigma(A) \geq \sigma(B \cup \{v\}) - \sigma(B)$$
$$A \subseteq B, \quad v \notin B$$
- Greedy obtains a $(1 - 1/e)$ approximation for the exact objective.
- CELF uses diminishing returns to skip repeated marginal-gain evaluations.

Implemented Methods

- Baselines:
  - random seed selection;
  - weighted out-degree;
  - PageRank.
- Centralized methods:
  - Monte Carlo greedy;
  - CELF lazy greedy.
- Distributed method:
  - partition nodes into m parts;
  - each worker sends at most q local candidates;
  - coordinator runs greedy on the merged candidate pool.

Experimental Setup

- Graphs:
  - Erdos–Renyi, $n = 40, 70$;
  - Barabasi–Albert, $n = 40, 70$;
  - Watts–Strogatz, $n = 60$;
  - Karate Club, $n = 34$.
- Seed budget: $k = 5$ or $k = 6$.
- Edge probabilities: weighted-cascade rule.
- Distributed sweep:

$$m \in \{2, 4, 8\}, \qquad q \in \{k, 2k, 5k\}.$$

Metrics

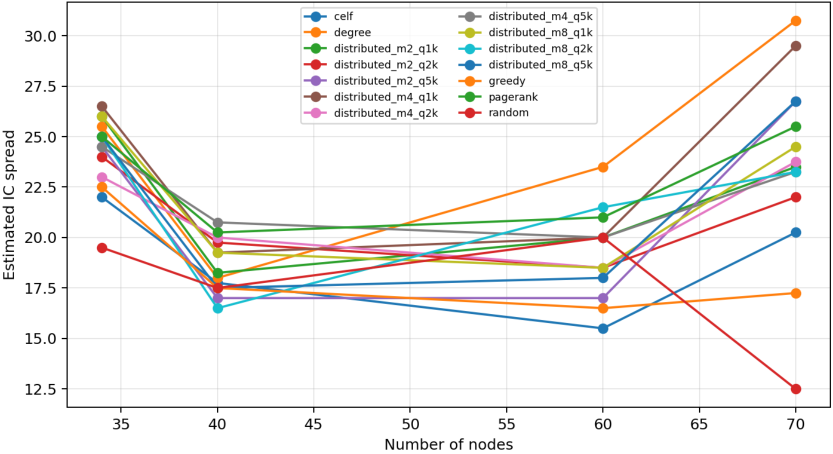
- Estimated influence spread.
- Runtime.
- Number of spread evaluations.
- Distributed quality ratio:
$$\rho = \frac{\sigma(S_{\mathrm{dist}})}{\sigma(S_{\mathrm{greedy}})}.$$
- Communication proxy: number of transmitted candidates.

Overall Results

Method	Spread	Ratio	Time	Eval.
Random	16.58	0.93	0.00	1.0
Degree	24.42	1.37	0.00	1.0
PageRank	22.92	1.27	0.00	1.0
Greedy	18.08	1.00	0.06	283.5
CELF	18.92	1.04	0.02	112.0
Best dist. avg.	24.00	1.34	0.32	359.5

- CELF reduces evaluations while keeping similar quality to greedy.
- Degree/PageRank are strong on these small weighted-cascade graphs.

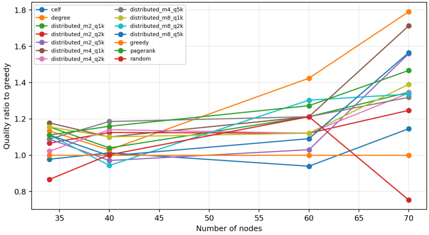
Influence Spread

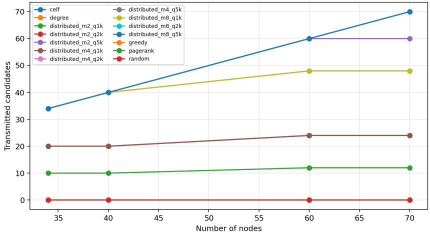


Distributed Tradeoff

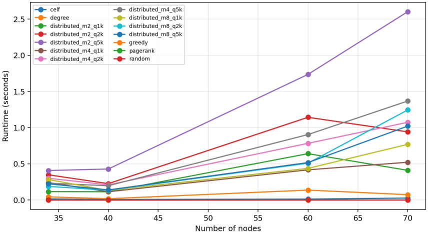
$(m, q/k)$	Spread	Ratio	Time	Eval.	Cand.
(2, 1)	21.58	1.19	0.30	321.5	11.0
(2, 2)	21.00	1.16	0.64	591.5	22.0
(2, 5)	21.50	1.20	1.37	1021.5	49.0
(4, 1)	24.00	1.34	0.32	359.5	22.0
(4, 2)	21.50	1.18	0.61	599.8	43.0
(4, 5)	22.08	1.22	0.71	683.8	52.3
(8, 1)	22.00	1.21	0.42	430.5	43.0
(8, 2)	21.00	1.16	0.58	503.2	52.3
(8, 5)	21.92	1.22	0.51	503.2	52.3

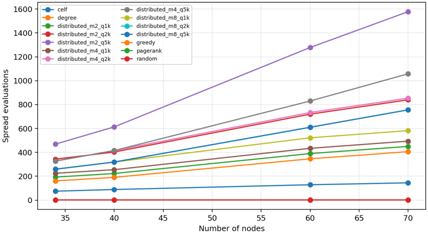
Quality and Communication





Runtime and Evaluation Counts





Interpretation

- Greedy theory applies to the exact submodular objective.
- Our implementation uses Monte Carlo estimates during selection.
- With small graphs and modest simulation counts, the selected seed set is noisy.
- This explains why degree/PageRank and some distributed settings can outperform greedy after fair re-evaluation.
- The experiment still reproduces the main algorithmic mechanisms:
  - submodular greedy selection;
  - lazy marginal-gain reuse;
  - communication-limited candidate sharing.

Limitations and Extensions

- Current graphs are smaller than the proposal's full scalability target.
- Monte Carlo simulation count is limited for runtime.
- Future improvements:
  - run more random seeds and report confidence intervals;
  - test Wiki-Vote or NetHEPT;
  - add RR-set based maximum coverage;
  - parallelize distributed worker computation.

Conclusion

- Influence maximization connects modern network applications to classical approximation algorithms.
- IC spread is monotone submodular, enabling greedy approximation.
- CELF demonstrates how theory becomes practical speedup.
- GreeDI-style candidate sharing highlights the cost of limited communication.
- Empirical quality depends on graph structure and spread-estimation accuracy.